

Chapter-15

SURFACE TENSION

- Properties of liquid due to which surface area minimizes.

- Factor affecting surface tension:

- Those factors which increases the cohesive force between molecules increases the surface tension.
- with increase in temperature surface tension decreases, at critical temp. it become zero.
- If impurity is completely soluble surface tension increases.
- If impurity is partly soluble it decreases the surface tension.
- On contamination surface tension decrease
- On electrification surface tension decreases.

- Surface tension is measured force per unit length in the plane of the liquid surface acting on imaginary line drawn in that surface.

$$T = \frac{F}{L}, \quad \text{unit} = \frac{N}{m}$$

$$[T] = MT^{-2}$$

- If a thread or wire of length ' ℓ ' floats on liquid surface then total length in contact with liquid surface is ' 2ℓ '.

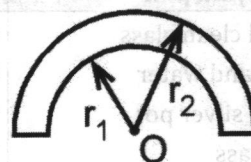
$$F = T \times 2\ell$$

- If a ring of negligible thickness and radius ' r ' floats on liquid surface the, surface tension force, $F = J \times 2 \times (2\pi r)$
- The pressure on **concave side** of curved surface is greater than **convex side**.
- Excess pressure is the pressure difference.
- Excess pressure on curved surface

$$P = T \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\text{If } R_1 = R_2 = R_1 \quad P = \frac{2T}{R}$$

- If curvature are in same direction excess pressure,



$$P = T \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

Case	Liquid drop in Air	Air bubble in liquid	Soap bubble
Fig.			
Free surface	1	1	2
Excess pressure	$\frac{2T}{R}$	$\frac{2T}{R}$	$\frac{4T}{R}$
Total inside pressure	$P_0 + \frac{2T}{R}$	$P_0 + \frac{2T}{R}$	$P_0 + \frac{4T}{R}$

$$P_{\text{exc}} \propto \frac{1}{r}; V \propto r^3 \quad \text{i.e., } V \propto \frac{1}{(P_{\text{ex}})^3}$$

$$\frac{P_1}{P_2} = \frac{r_2}{r_1}, \quad \frac{V_1}{V_2} = \left(\frac{P_2}{P_1} \right)^3$$

- When two soap bubble coalesce, then the radius of curvature of common surface.

$$r = \frac{r_1 r_2}{r_2 - r_1} \quad \text{where } r_1 < r_2$$

- When two soap bubble of radii r_1 and r_2 combine to form a new bubble in vacuum under isothermal condition, its radius is given by

$$r = \sqrt{r_1^2 + r_2^2}$$

- If P is the pressure V and A are the change in volume and surface area when two bubble coalesce in vacuum under isotherm condition them.

$$3PV + 4AT = 0$$

Fig.

$F_A V / S F_C$	$F_A > \frac{F_C}{\sqrt{2}}$	$F_A < \frac{F_C}{\sqrt{2}}$	$F_A = \frac{F_C}{\sqrt{2}}$

Surface Tension

Angle of contact	$\theta < 90^\circ$	$\theta > 90^\circ$	$\theta = 90^\circ$
	Meniscus is concave	Meniscus is convex	Meniscus is plane
	Liquid rises	Liquid falls	Neither rise nor fall
	Liquid wet solid surface	Liquid do not wet solid surface	Does not wet capillary tube only touches

Solid-liquid pair	Angle of contact
(i) Pure water and clean glass	0°
(ii) Ordinary glass and water	8°
(iii) Water and silver pot	90°
(iv) Mercury and glass	135°
(v) Alcohol and glass	0°

Factors affecting angle of contact

- The nature of solid and liquid in contact
- The given pair of liquid and the solid
- the impurities
 - With increasing temperature, angle of contact decrease.
 - If $T \uparrow$ sec, angle of contact also increases
 - Height raised in capillary tube is given by

$$h = \frac{2T \cos \theta}{r \rho g} \Rightarrow h \propto \frac{1}{r}$$
 - $hr = \text{constant} \Rightarrow$ known as Jurin's law
 - If a capillary tube is dipped into a liquid and tilted at an angle ' α ' with vertical, then vertical height of liquid remains same but length of liquid column increase as, $\ell = \frac{h}{\cos \alpha}$.
 - If insufficient length of capillary tube is dipped in liquid then liquid do not overflow but radius of meniscus increases.
 - Inside a satellite, water will rise up to top level but do not come outside.
 - Oil rises in a lamp wick is due to capillary action.
 - Work done in splitting of a bigger drop into smaller drop:
 - If a bigger drop of radius 'R' splits into 'n' equal smaller drops of radius 'r' then, work done is
 - $W = T \cdot \Delta A$

$$W = 4\pi R^2 T \left(\frac{1}{n^{\frac{1}{3}}} - 1 \right)$$

- In this process temperature decreases,
- Energy is absorbed
- Decrease in temperature ($\Delta \theta$)

$$= \frac{3T}{J \rho S} \left(\frac{1}{r} - \frac{1}{R} \right)$$

- When 'n' smaller drops each of radius 'r' are combined to form a bigger drop of radius 'R' then,

$$W = 4\pi r^2 T \left(n - n^{\frac{2}{3}} \right)$$

- Energy is released
- Temperature increases
- $\Delta \theta = \frac{3T}{J \rho S} \left(\frac{1}{r} - \frac{1}{R} \right)$
- Force required to separate, two parallel plates of area 'A' and 'd' is the distance between two plates,

$$F = \frac{2T}{d} A$$
- Charging of bubbles increase surface area.

Multiple Choice Questions

- A soap bubble (surface tension $30 \times 10^{-3} \text{ N/m}$) has a radius 2 cm. The work done in doubling the radius is
 - $1.1 \times 10^{-4} \text{ J}$
 - $2.2 \times 10^{-4} \text{ J}$
 - $9.0 \times 10^{-4} \text{ J}$
 - zero
- For liquid & glass angle of contact value is 0° , if glass tube is inserted in liquid, liquid rises up to 6 cm height if other capillary tube has internal radius $3/2$ times of previous, then liquid column height will be
 - 9 cm
 - 3 cm
 - 4 cm
 - 12 cm
- The most characteristic property of a liquid is
 - elasticity
 - fluidity
 - formlessness
 - volume conservation
- Soap helps in better cleaning of clothes because
 - It reduces the surface tension of solution
 - It gives strength to solution
 - It absorbs the dirt
 - Chemical of soaps change
- Two capillary tubes P and Q are dipped in water. The height of water level in capillary P is $2/3$ the height in Q capillary. The ratio of their diameters is
 - 2:3
 - 3:2
 - 3:4
 - 4:3

6. The surface tension of soap solution is $25 \times 10^{-3} \text{ Nm}^{-1}$. The excess pressure inside a soap bubble of diameter 1 cm is
- 10 Pa
 - 20 Pa
 - 5 Pa
 - None of the above
7. Two bubble of radii 3cm and 5cm radii coalesce to form a common surface, the radius of curvature of this common surface will be
- 3cm
 - 5cm
 - 20 cm
 - 7.5cm
8. The work done in splitting a drop of water of 1mm radius into 10^6 droplets is (S.T. of water $72 \times 10^{-3} \text{ J/m}^2$)
- $9.98 \times 10^{-5} \text{ Joule}$
 - $8.95 \times 10^{-5} \text{ Joule}$
 - $5.89 \times 10^{-5} \text{ Joule}$
 - $5.98 \times 10^{-5} \text{ Joule}$
9. Water rises to a height of 10 cm in a capillary tube, and mercury falls to a depth of 3.42 cm in the same capillary tube. If the density of mercury is 13.6 and the angle of contact is 135° , the ratio of surface tension for water and mercury is
- 1 : 05
 - 1 : 3
 - 1 : 6.5
 - 1.5 : 1
10. A vessel, whose bottom has round holes with diameter of 0.1 mm, is filled with water. The maximum height to which the water can be filled without leakage is (S.T. of water = 75 dynes/cm, $g = 1000 \text{ cm/s}^2$)
- 100cm
 - 75 cm
 - 50 cm
 - 30 cm
11. Water rises in a capillary tube when its one end is dipped vertically in it is 3 cm. If the surface tension of water is $75 \times 10^{-3} \text{ N/m}$, then the diameter of capillary will be
- 0.1 mm
 - 0.5 mm
 - 1.0 mm
 - 2.0 mm
12. If the work done in blowing a bubble of volume V is W then the work done in blowing a soap bubble of volume $2V$ will be:
- W
 - $2W$
 - $\sqrt{2} W$
 - $4^{1/3} W$
13. In a capillary tube water rises to a height 16 cm. The rise of water in a capillary tube of radius double to that of previous one is
- 16 cm
 - 8 cm
 - 32 cm
 - 64 cm
14. On dipping a capillary in a liquid the liquid rises to a height of 3 cm in it. Now the capillary is dipped in such a way that only a part of it of length 1 cm remains out side the liquid then
- Liquid will come out in form of spray from capillary
 - Liquid will leak form capillary
 - Liquid will rise up to end & radius of curvature of liquid surface gets change
 - Liquid will rise upto end & radius of curvature of liquid surface remain unchanged
15. On dipping a capillary in a liquid inside an artificial satellite
- the shape of the liquid meniscus is concave
 - the shape of the liquid meniscus is convex
 - the liquid will not rise at all
 - the liquid will rise upto the length of the tube
16. The surface tension of soap is T . The work done in blowing a soap bubble of diameter D to that of a diameter $2D$ is -
- $2\pi D^2 T$
 - $4\pi D^2 T$
 - $6\pi D^2 T$
 - $8\pi D^2 T$
17. A square frame of side L is dipped in a liquid soap. When it is taken out of the liquid a soap film is formed on it. If the surface tension of soap is T , then the force acting on frame is
- $2TL$
 - $4TL$
 - $8 TL$
 - $10 TL$
18. The work done in blowing a bubble of radius R is W , then the work done in making a bubble of radius $2R$ from that solution is
- $W/2$
 - $2W$
 - $4W$
 - $2^{1/3} W$

Answers:

1. c	2. c	3. d	4. a	5. b
6. b	7. d	8. b	9. c	10. d
11. c	12. d	13. b	14. c	15. d
16. c	17. c	18. c		

Solution for Selected Problems

- $$W = T \Delta A = 30 \times 10^{-3} [4\pi \times (2r)^2 - 4\pi r^2] \times 2$$

$$= 130 \times 10^{-3} \times 12 \times \pi \times (2 \times 10^{-2})^2 \times 2$$

$$= 9.0 \times 10^{-4} \text{ J}$$
- $$h = \frac{2T \cos \theta}{r \rho g}$$

$$\Rightarrow h \propto \frac{1}{r} \Rightarrow hr = \text{constant}$$

$$h_1 r_1 = h_2 r_2$$

$$\Rightarrow 6 \times r = h_2 \times \frac{3}{2} r$$

$$\therefore h_2 = 4 \text{ cm}$$

$$5. h_1 r_1 = h_2 r_2$$

$$h_p \times r_p = h_q \times r_q$$

$$\Rightarrow \frac{r_p}{r_q} = \left(\frac{h_q}{h_p} \right) = \frac{3}{2}$$

$$\text{Ratio of diameter} = \frac{D_p}{D_q} = \frac{r_p}{r_q} = \frac{3}{2}$$

$$6. T = 25 \times 10^{-3} \text{ N.m}$$

$$d = 1 \text{ cm}$$

$$r = 0.5 \times 10^{-2} \text{ m} = 5 \times 10^{-3} \text{ m}$$

$$\text{Excess pressure inside } P = \frac{4T}{r} = \frac{4 \times 25 \times 10^{-3}}{5 \times 10^{-3}}$$

$$P = 20 \text{ N/m}^2 = 20 \text{ Pa}$$

$$7. \text{ Common radius,}$$

$$r = \frac{r_1 r_2}{r_2 - r_1} = \frac{3 \times 5}{5 - 3} = \frac{3 \times 5}{2} = 7.5 \text{ cm}$$

$$8. T = 72 \times 10^{-3}$$

$$n = 10^6$$

$$R = 1 \text{ mm} = 10^{-3} \text{ m}$$

$$W = T \cdot \Delta A = T \times 4\pi R^2 [n^{1/3} - 1]$$

$$= 72 \times 10^{-3} \times 4 \times \pi \times 10^{-6} \times (10^2 - 1)$$

$$= 8.95 \times 10^{-5} \text{ J}$$

$$9. h = \frac{2T \cos \theta}{r \rho g}$$

$$T = \frac{h r \rho g}{2 \cos \theta}$$

$$\Rightarrow \frac{T_{\text{water}}}{T_{\text{mercury}}} = \frac{h_w \times \rho_w \times \cos 135^\circ}{h_{Hg} \times \rho_{Hg} \times \cos 0^\circ}$$

$$\Rightarrow \frac{T_{\text{water}}}{T_{Hg}} = \frac{10 \times 1 \times (-1/\sqrt{2})}{(-3.42) \times (13.6) \times 1}$$

$$\therefore \frac{T_{\text{water}}}{T_{Hg}} = 1:6.5$$

$$10. h = \frac{2T \cos \theta}{r \rho g} = \frac{2 \times 75 \times 10^{-3} \times 1}{0.05 \times 10^{-3} \times 1000 \times 10}$$

$$h = 30 \times 10^{-2} \text{ m}$$

$$= 30 \text{ cm}$$

$$11. h = \frac{2T \cos \theta}{r \rho g} = \frac{4T \cos \theta}{d \rho g}$$

$$\Rightarrow d = \frac{4T \cos \theta}{h \rho g} = \frac{4 \times 75 \times 10^{-3} \times 1}{3 \times 10^{-2} \times 1000 \times 10}$$

$$\therefore d = 1 \text{ mm}$$

$$12. V = \frac{4}{3} \pi r^3$$

$$\Rightarrow r \propto v^{1/3}$$

$$\therefore w = 4\pi r^2 T$$

$$\Rightarrow w \propto r^2 \propto v^{2/3}$$

$$\Rightarrow \frac{w_2}{w_1} = \left(\frac{v_2}{v_1} \right)^{2/3} = 2^{2/3}$$

$$\therefore w_2 = 4^{1/3} w$$

$$13. h \propto \frac{1}{r}$$

$$\Rightarrow \frac{h_2}{h_1} = \left(\frac{r_1}{r_2} \right) = \left(\frac{r}{2r} \right)$$

$$\therefore h_2 = \frac{16}{2} = 8 \text{ cm}$$

$$16. w = T \times 2 \cdot \Delta A$$

$$= T \times 2 [\pi (2D)^2 - \pi D^2]$$

$$= 2T \times 3\pi D^2$$

$$= 6\pi D^2 T$$

$$17. F = T \times 2 \times 4L = 8TL$$

$$18. w \propto r^2$$

$$\frac{w_2}{w_1} = \left(\frac{r_2}{r_1} \right)^2 = \left(\frac{2R}{R} \right)^2$$

$$\therefore w_2 = 4w$$

